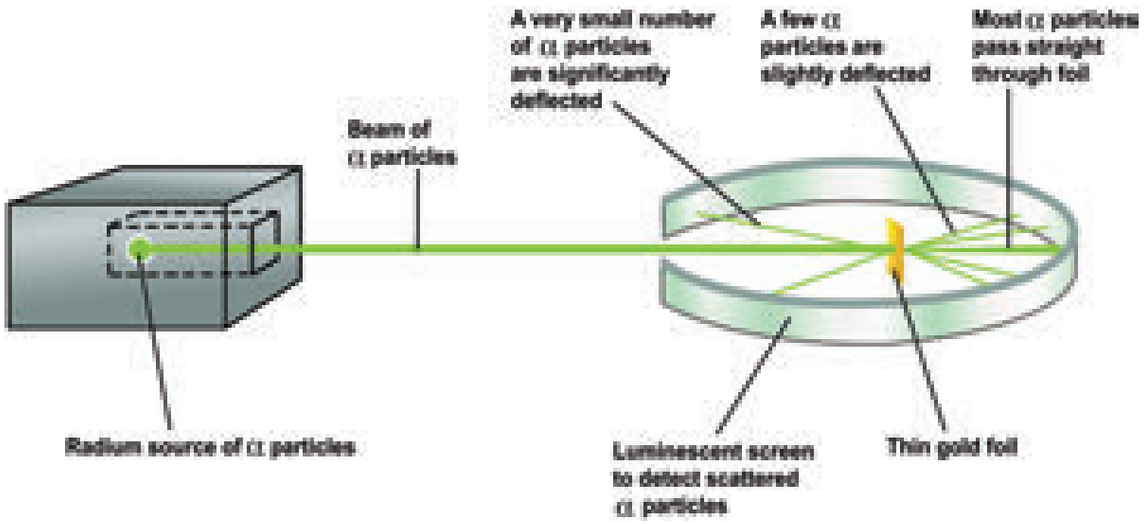


Figure 11.2 Deflection of α-particle by a gold leaf

Later, Rutherford generalized these results of alpha particles scattering experiment and suggested a model of the atom that is known as Rutherford's Atomic model.

11.1.1 Rutherford's Atomic model

According to this model :

i. The atom contains large empty space.

ii. There is a positively charged mass at the centre of the atom, known as nucleus.

iii. The size of the nucleus of an atom is very small compared to the size of an atom.

iv. The electrons revolve around the nucleus in close circular paths called orbits.

v. An atom as a whole is electrically neutral, i.e., the number of protons and electrons in an atom are equal.

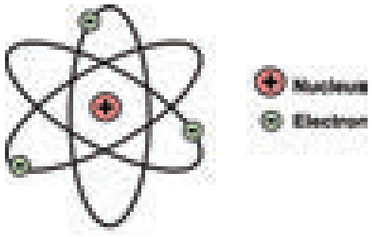


Figure 11.3 Rutherford's model of the atom was some what like that of the solar system.

Rutherford's model of atomic structure is similar to the structure of the solar system. Just as in the solar system, the Sun is at the centre and the planets revolve around it, similarly in an atom the nucleus present at the centre and the electrons revolve around it in orbits or shells.

11.1.2 Limitations in Rutherford's model

According to Electromagnetic theory, a moving electron should accelerate and continuously lose energy. Due to the loss of energy, path of electron may reduce and finally the electron should fall into the nucleus. If it happens so, atom becomes unstable. But atoms are stable. Thus, Rutherford's model failed to explain the stability of an atom.

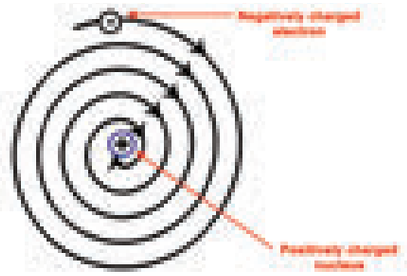


Figure 11.4 Showing an atom losing energy.

11.2 Bohr's model of an atom

In 1913, Neils Bohr, a Danish physicist, explained the causes of the stability of the atom in a different manner. The main postulates are:

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